INDEX

NAME: Abiramii PL SUBJECT: Data Mining

STD. _____ DIV. _____ ROLL NO. 220701007 SCHOOL _____

SR. NO.	PAGE NO.	TITLE	DATE	TEACHER'S SIGN / REMARKS
1.	20/12/25	Instaling weka & Exploring a dataset.	~~20/12/25~~	
2.	27/12/25	Loading a dataset & visualize the data.		
3.	5/1/26	Classifier algoritems Run decision trees		
4.	10/1/26	Forming Clusters: Run clustering algorithm		
5.	24/1/26	Mining Association Rules - Run Apriori algorithm.		
6.	31/1/26	Exploring Numpy & Pandas Package		
7.	7/2/26	Data wrangling using Pandas		
8.	14/2/26	Data Manpulation using Pandas		
9.	21/2/26	Linear Regression Implementation		
10.	28/2/26	K-means Implementation.		

Completed 28/3/26

20/3/25

Experiment-1 - Installing WEKA and Exploring a dataset.

Aim:

To install the WEKA ted and explore datasets for basic data mining operations.

Procedure:

1. Open a web browser and download WEKA. from basic data mining operations download datasets.
2. Install software following installation instructions.
3. Launch WEKA applications and select WEKA explorer interface.
4. Click open file and load sample datasets.
5. View attributes and instances in the preprocess tab.
6. Observe summary statistics such as mean, minimum, maximum.

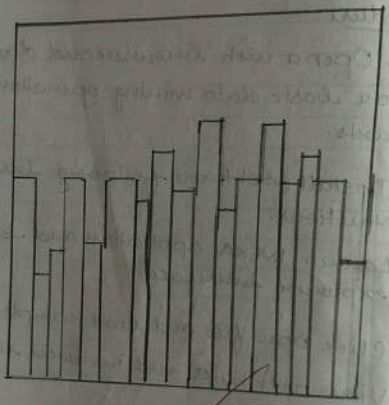
Output:

Dataset successfully loaded showing attribute details.

Result:

WEKA was installed and dataset exploration was completed successfully.

Visualisation:



8/10

27/12/25 Experiment-2: Loading a dataset and visualize the data.

Aim:

To visualize data using WEKA graphical tools.

Procedure:

1) Launch WEKA explorer.
2) Click open File to import dataset.
3) Go to pre-process tab to confirm loading.
4) Select visualize tab.
5) Choose attribute pairs for plotting.
6) Enable class coloring option.
7) Observe scatter plot output.

Output:

Visual plots generated showing attribute relationships.

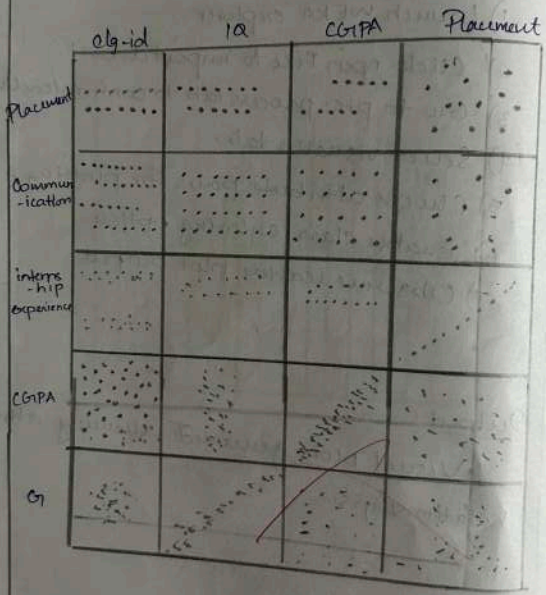
Result:

Data visualization was successfully performed.

3/1/26

Output:

Plot matrix



8/10

3/1/26 Experiment-3: Build a classification Run decision Tree, Naive bayesian classifier, NN classifier & SVM.

Aim:

To apply classification techniques using WEKA.

Procedure:

1) Load dataset in WEKA Explorer.
2) Select classify tab.
3) Choose classifier J48 (decision tree).
4) Set evaluation method to cross-validation
5) Click start & note results.
6) Repeat Naive-bayes, NN, SVM.
7) Compare accuracy of all classifiers.

Output:

Performance metrics displayed for each algorithm.

Result:

Classification models were successfully built & compared.

Classifier output:

Scheme: weka.classifiers.trees.J48 -C 0.25 -M 2

Relation: college

Instances: 1000

Attributes: 10
- collegeID
- IQ
- CGPA
- communication-skills
- Placement
- Internship-Experience

Test mode: 10-fold cross-validation

Classifier model (full training set)

J48 pruned tree

Communication-skills <= 7

IQ <= 110: No (529.0)

IQ > 110

CGPA <= 8: No (981.0)

CGPA > 8

CGPA > 8

Projects-completed <= 1

IQ <= 110: No (309.0)

IQ > 110: yes (78.0)

Projects-completed > 1: yes (865.6)

Number of leaves: 10

Size of the tree: 19

== Confusion matrix ==

a	b	← Classified as
8341	0	a = No
0	1659	b = yes

9/10

10/1/26

Ex: 4 - Forming clusters: Run-clustering - algorithms.

Aim:

To perform clustering on a given dataset using a clustering algorithm & group similar data points into clusters.

Procedure:

1) Import the require libraries.
2) Load the dataset into WEKA-System.
3) Preprocess the data
4) Apply the clustering algorithms to the dataset.
5) Run & fit the model & assign cluster labels to data points.
6) Visualize the clusters.

Output:

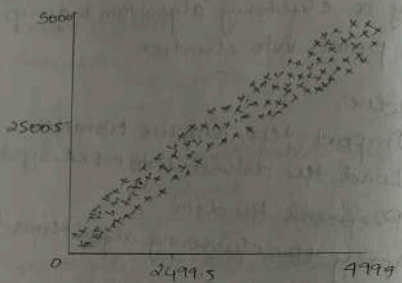
- data points are grouped into distinct clusters.
- A graphical representation of clustered data points with different colors.

Result:

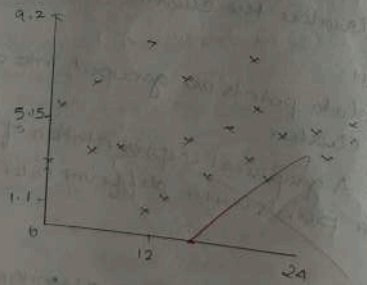
Thus the clustering algorithm was successfully executed and the datasets was divided into meaningful clusters based on similarity.

Output:

Simplelinear cluster:



Kmeans cluster:



23/10

Ex-5: Mining Association Rules-
Run Apriori algorithm

Aim:

To generate frequent itemsets and strong association rules from a transaction dataset using the apriori algorithm.

Procedure:

1) Open weka & go to the preprocess tab and load the dataset.

2) Ensure that the dataset contains nominal attributes suitable for association rule mining.

3) Go to the association tab & click choose & select Apriori algorithm from the list.

4) Set minimum support & confidence values if required.

5) Observe the generated frequent itemsets & association rules in the panel.

Output:

List of frequent itemsets with their support values.

Result:

Thus the apriori algorithm was successfully implemented in WEKA to generate frequent itemset and discover strong association rules from the given dataset.

Output:

=== association model ===

Apriori

minimum support: 0.05 (instances)

min metric <confidence>: 0.9

generate sets of large itemsets:

size of set of large itemsets L(1): 9

size of sets of large itemsets L(3): 4

Best Results Found:

1. T3=Man2 => T1=A2 <conf:(1) lift:(1.33) lev:(0.13) conv:(0.5)

2. T2=B1 => T1=A 1. <conf:(1)> lift:(1.3) lev:(0.08) [0] conv:(0.25)

3. T2=D 1 => T2=A1 <conf:(1)> lift (4) lev: (0.19) conv:(0.75)

31/1/26

8/10

31/1/26

Exp: 6: Exploring Numpy & Pandas Package

Aim:

To explore and perform basic ~~oo~~ operations using Numpy and Pandas Dataframes.

Procedure:

1. Import numpy & pandas library
2. Create Numpy arrays & Generate multi-dimensional arrays.
3. Create a Pandas dataframe, operations & perform basic Dataframe operations.
4. Access & modify rows & columns.
5. Display results.

Output:

Numpy array Operations
Display dataframe
Statistical summary

Result:

Thus the basic functionalities of Numpy & Pandas were successfully explored & implemented.

Output:

Data frame:

	Name	Age	Marks
0	Mukesh	21	85
1	Arun	22	90
2	Ravi	23	88

Statistical summary:

	Age	Marks
Count	22.0	87.667
Mean	1.0	2.516611
Std	21.5	86.500
min	23.5	88.000
25%	22.5	89.000
75%	22.5	89.000
max	23.5	90.000

9/10

20/3

7/2/26 Exp:7 - Data wrangling using Pandas

Aim:

To perform data cleaning & preprocessing using Pandas.

Procedure:

1. Import pandas library
2. Load dataset using read_csv()
3. Identify missing values using isnull()
4. Handle missing data using fillnull or dropnull
5. Rename duplicate records & columns.
6. Save the cleaned dataset.

Output:

- Cleaned dataset with handled missing values.
- Updated dataframe after removing duplicates.

Result:

Thus, data arranging was successfuly performed by cleaning & preprocessing the dataset.

Output:

Original data:

	Name	Marks	City
0	A	85.0	Chennai
1	B	Nan	Delhi
2	C	90.0	Mumbai

Missing values:

	Name	Marks	City
0	False	False	True
1	False	~~False~~ True	False
2	False	False	False

Cleaned data:

	Name	Marks	City
0	A	85.00	Chennai
1	B	85.057	Delhi
2	C	90.000	Mumbai

9/10

4/2/26

4/2/26

Exp: 8: Data Manipulation Using Pandas

Aim:

To perform Data manipulation techniques using Pandas.

Procedure:

1. Import Pandas library
2. Sort data using sort_values()
3. Filter records using conditions
4. Group data using groupby()
5. Perform aggregation functions (mean, sum, count)
6. Display manipulated data.

Output:

- Sorted dataset
- Filtered dataset.
- Grouped data with aggregated results.

Result:

Thus, the data manipulation operations were successfully performed using Pandas functions.

Output:

Sorted data:

	Name	Dept	Marks
0	A	IT	85
1	B	CSE	90
2	C	IT	88
3	D	ECE	75

Filtered data (Marks ≥ 80)

	Name	Dept	Marks
0	A	IT	85
1	B	CSE	90
2	C	IT	88

Average Marks by dept:

Dept	Marks
CSE	92.5
ECE	75.0
IT	86.5

10/10
22/3

Exp-9: Linear Regression Implementation

Aim:

To implement linear regression using Python & Predict dependent variable values.

Procedure:

1. Import required libraries & load dataset
2. Separate independent & dependent variables.
3. Split dataset into training & testing sets.
4. Apply linear regression model.
5. Fit the model on training data.
6. Predict output & evaluate model performance.

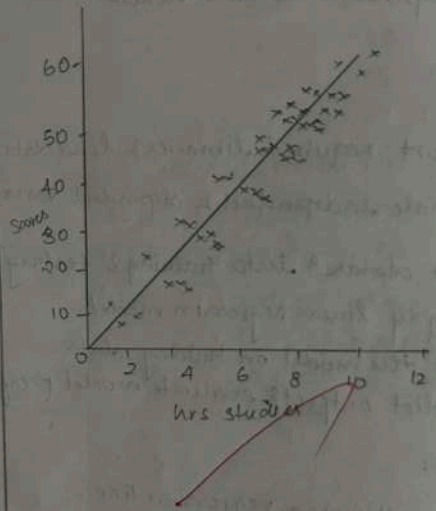
Output:

Graph showing regression line.

Result:

Thus linear regression was successfully implemented & the relationship between variables analyzed.

Output:



28/10/24

Exp 10: K-means Implementation

Aim:

To implement k-means clustering algorithm & group similar data points into clusters.

Procedure:

1. Import required libraries & load dataset.
2. Apply k-means alg. & fit model to dataset.
3. ~~Apply~~ visualize clusters using scatter plot.

Output:

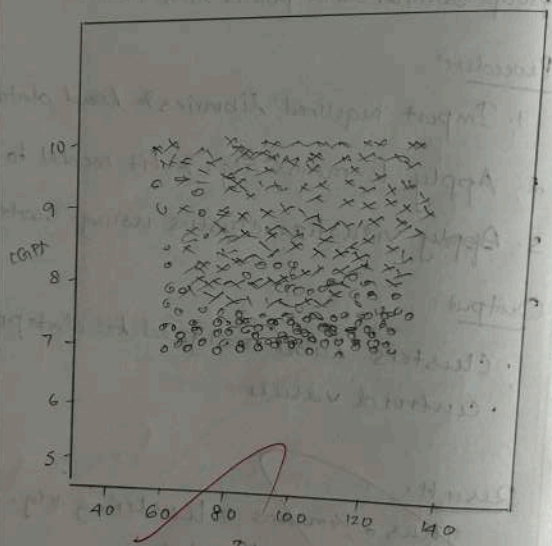
- Clusters label assigned to data points
- centroid values.

Result:

Thus, Kmeans clustering alg. was successfully implemented.

Output:

K-means clustering



20/3 10/10